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challenge transforms into a catch-22 in the case of embedded systems—the security algorithms need to be very comprehensive because of the real-time nature of embedded system operations; yet they need to be small and low-power.

Prof. Anupama explains, “Embedded devices are low-power, small-sized and energy-hungry devices. The security algorithms available in the market are processor hungry, which contradicts the basic axioms of the embedded devices. The current encryption and decryption algorithms are complex and require high processing power, and hence cannot be implemented easily on the low processing capability embedded devices. Making the embedded devices complex is not a solution as this will only increase the price of the devices—easy-to-implement encryption and decryption algorithms need to be developed.”

To overcome these challenges and incorporate the required levels of security consistent analysis and audits throughout the product engineering lifecycle would be required. It would require the setting up of a holistic security mechanism, working from the design/architecture upwards.

We have come a long way

A path-breaking story titled ‘*The Embedded Internet*’ was published in the *Wired* magazine in 1996. The article introduced a miraculous technology that could change the face of the Internet, and the world—“Tiny crash-proof computers that are embedded or hardwired within everyday products and dedicated to the performance of specific tasks or groups of tasks.” No prizes for guessing that the revolutionary technology discussed therein is nothing but the humble embedded system that is so ubiquitous today.

Read the article at http://www.wired.com/wired/archive/4.10/es_embedded_pr.html to realise how far we have come in the past decade! What the author called exciting new stuff then is commonplace now; things he called dreams are a reality today; but the ultimate goal of an “Internet of Things” remains to be realised!

Other challenges...

Security is just one of the many challenges in the making of embedded Internet systems. Some of the other issues include:

Power management. Current embedded devices employ high-end peripherals and heavy software stacks leading to the increased consumption of power, and consequent reduction in battery life. Employing peripherals that support low power modes is one of the tried and tested mechanisms for overcoming this issue, but there is a lot of scope for improvement in this area. “Power management also requires focus on the system architecture that includes hardware as well as software,” says Sable.

Full usage of channel. Most of the embedded devices available are capable of supporting communication over fast ethernet or even gigabit ethernet. “Though the communication is over the ethernet, they are not capable of using the channel to the full extent. On an average, most of the embedded systems are capable of transferring only a few bytes of data while meeting the energy constraint of the applications in parallel,” says Prof. Anupama. Hence, in the future, engineers will have to work out how to enable the embedded Internet devices to make full use of the channel, without considerable increase in power consumption.

Predictability of delays. Current Internet technologies do not feature predictable delays in data delivery. Most Internet-based embedded applications are real-time, and hence predictable time delay is a basic requirement. Although modifications such as the real-time

Ethernet exist, they are yet to be implemented in all embedded devices.

Need for better middleware. The multitude of embedded devices and the variations in hardware and software configurations of these devices require the use of middleware—an area still under development.

Fortunately, there is loads of motivation to solve these and other issues that remain in the embedded Internet system space, since the demand for such systems is growing phenomenally and expected to grow even faster. With the projected growth of cloud computing, e-commerce and other applications in the near-future, as well as the enthusiasm to implement the Internet of Things, the demand for Internet-capable embedded systems is only likely to see an upwards trend.

“Embedded computing is undergoing a transformation. Breakthroughs in computing per watt, in parallel with advances in wired and wireless broadband connectivity, will enable the development of billions of new intelligent embedded devices. We foresee thousands of devices embedded in civil infrastructure (buildings, bridges, water ways, highways and protected regions) to monitor structural health and detect crucial events. Eventually, such devices might be tiny enough to pass through bodily systems or be usable in large enough numbers to instrument major air or water flows! In short, the demand for such systems is expected to grow rapidly,” signs off Prof. Anupama. ●

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